Part II, Lecture 3: Severity, Exposure, Expected and Unexpected Loss, VaR, and the Two Risk Paradigms

By the end of this lecture you must know how the loss magnitude side of credit risk works (LGD and EAD), how expected loss rates aggregate from one credit line up to the whole portfolio, why expected loss is covered by provisions while unexpected loss requires capital, why standard deviation is the wrong variability measure for credit losses and VaR is the right one, how diversification makes portfolio VaR smaller than the sum of individual VaRs, and how the Default Mode and Mark-to-Market paradigms differ, including migration risk, spread risk and pro-cyclicality.

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1. From Default Probability to a Complete Loss Measure

Lectures 1 and 2 built the probability side: ratings produce PDs. But a default probability alone says nothing about how much money is actually lost when default happens. Two exposures with the same PD can carry very different risk, because the amount lost per euro at risk and the number of euros at risk at the moment of default both vary. This lecture completes the picture.

Expected loss for a single exposure (simplified): EL = PD x LGD x EAD, where PD is the probability of default over the horizon, LGD the share of exposure lost if default occurs, and EAD the amount outstanding at the moment of default.

2. Loss Given Default: Two Estimation Approaches

LGD measures severity: the amount lost if default occurs. It is the mirror of the recovery rate.

LGD = 1 - RR, where RR is the recovery rate. Recovery varies hugely with collateral, guarantees and the legal process, which is why LGD deserves its own estimate rather than a flat assumption.

The course gives two estimation approaches.

- **Actual payouts approach**, used for loans on the basis of historical recovery data: discount the net recoveries (recovery value RV minus recovery costs RC, at each date t, at rate i) and compare with the exposure at default:

LGD = 1 - RR = 1 - [Sum over t of (RVt - RCt) / (1+i)^t] / EAD

Key inputs: recovery value, recovery costs, timing, the time value of money, and EAD. The timing term is not cosmetic: the slides show that the average months needed to execute a bankruptcy procedure differ widely across Italian courts, so geography itself moves LGD through the discounting.

- **Market price approach**, used for traded bonds: look at market prices of similar bonds that defaulted in the past, usually observed **one month after default**. The delay lets the immediate default-driven price volatility subside, so the price better reflects recovery expectations:

LGD = 1 - Market Value (1 month post-default) / Nominal Value

The average LGD of comparable bonds (matched on seniority, debtor location, collateral and other relevant features) becomes the ex ante estimate for currently traded bonds.

3. Expected Loss Rate: One Line, One Borrower, the Whole Portfolio

Once PD and LGD are estimated, the expected loss rate of a specific credit line is a simple product. Aggregation then proceeds in two steps, each weighted by exposure.

Per credit line: ELRij = PDi x LGDij, where i is the obligor and j the specific credit line of that obligor. One borrower keeps a single PD across lines; severity differs line by line.

Per borrower with m credit lines: ELRi = PDi x Sum over j of [LGDij x (EADij / EADi)], a weighted average of line-level LGDs with weights given by each line's share of the borrower's total exposure at default.

Portfolio of n borrowers: ELRp = Sum over i of [ELRi x EADi / (Sum over i of EADi)]. And ex post, the same weighting applied to realised loss rates gives the portfolio loss rate LRp.

Notice what changes at each level: the PD belongs to the borrower, the LGD to the individual line, and only exposures provide the weights. Mixing these levels up is the most common mechanical error with these formulas.

4. Exposure at Default and the Exposure Multiplier

EAD is the outstanding amount at the moment of default, and it is often **not** the current exposure: borrower behaviour changes as default approaches.

EAD = Em x Ec, where Ec is the current exposure and Em the exposure multiplier, estimated from historical data. For each past defaulter the bank computes Em = Actual EAD / Ec one year before default, then clusters similar loans (same facility type, similar size) and averages Em within each cluster.

- **Discretionary facilities (credit lines, cards)**: borrowers draw down before default, so typically Em > 1 and EAD exceeds the current exposure.
- **Amortizing loans (mortgages, term loans)**: scheduled repayments reduce principal before default, so typically Em < 1 and EAD is below the current exposure.

With all three components in place, expected loss in value for a borrower is ELi = PDi x Sum over j of (LGDij x EADij), and the portfolio expected loss is simply the sum over borrowers: ELp = Sum of ELi. Ex post, portfolio loss is the sum of the actual losses on the individual exposures.

5. Expected Loss in Value, Provisions, and the Definition of Default

EL is an expected cost, and as such it is covered by **provisions** in the bank's financial statements. What counts as a default therefore moves every number in this lecture.

- **Traditional definition**: focused on bad loans, where the bank evaluates that the borrower will not pay back in full, without considering forced execution of collaterals and guarantees.
- **Basel II definition of default**: broader. It includes exposures more than **90 days past due** and borrowers classified **Unlikely to Pay (UTP)**.

The conceptual advancement: a wider spectrum of credit deterioration is captured, and objectivity increases. The 90-days-past-due leg derives from pure observation of data, whereas UTP is more judgmental and can be interpreted differently by banks and supervisors around the world. Widening the definition raises measured PDs and changes the meaning of LGD and EAD estimates built on it.

6. Expected versus Unexpected Loss

Expected loss: the average loss over a medium-long period. Predictable, recurring, covered by provisions.

Unexpected loss: actual losses deviate from the average over short periods. This variability around the mean requires risk capital.

The lecture's roulette analogy fixes the intuition: betting always and only on red has a negative expected outcome, yet any single spin can still land far from that average. Managing a bank means provisioning for the expectation and capitalising for the deviation, so that one bad year does not sink the institution. The new goal of the lecture is to measure and manage UL.

7. Why Standard Deviation Fails and VaR Takes Over

The natural variability measure, standard deviation, is symmetric: it treats upside and downside deviations alike. Credit loss rate distributions are highly **skewed**: many outcomes sit near zero loss on the left, with a long tail of potentially high losses on the right. A symmetric measure does not focus on the critical bad tail.

Value at Risk estimates the maximum loss expected at a given confidence level over a set horizon, excluding EL (net VaR). A 1-year VaR at 99% confidence means: 99% probability that actual losses will NOT exceed EL + VaR, and a 1% probability that they will.

- VaR is the **economic capital** needed to absorb unexpected losses. If actual loss exceeds EL (provisions) plus VaR (capital), the bank faces distress or failure.
- The confidence level reflects the bank's target probability of default from credit losses: a 99.9% VaR leaves a 0.1% chance that losses exhaust the capital buffer. The cost of this capital feeds loan pricing and risk-adjusted performance measures.

8. Portfolio Effects: Diversification, Concentration, Marginal VaR

EL is additive: ELp = Sum of ELi.

VaR is NOT additive: portfolio VaR is lower than the sum of the stand-alone VaRs, because defaults are imperfectly correlated. The key variable is the correlation between losses on individual exposures.

- **Concentration risk** is the risk of a sub-optimally diversified portfolio: exposures highly correlated through a shared industry, region or risk factor. High concentration means higher portfolio VaR and more capital required.
- **Marginal VaR** measures what actually matters for a decision: the incremental risk a loan adds to the existing portfolio, VaR(i,M) = VaR(p+i) - VaR(p). It is generally lower than the loan's stand-alone VaR because of the diversification benefit inside the portfolio.

Variability concerns not only losses as a whole but also their components: default rates, loss given default rates, and exposure multipliers all fluctuate, and full models account for each.

9. Credit Risk Models: SCRM versus FPCRM

Credit risk models estimate probability distributions of credit outcomes in order to compute VaR. The course distinguishes two families.

- **SCRM, simplified models**: Default Mode only, stand-alone VaR, no correlations.
- **FPCRM, full portfolio models**: portfolio and marginal VaR, correlations incorporated, either Default Mode or Mark-to-Market.

CRM versus ratings: ratings yield the EL inputs (PD and LGD estimates); credit risk models yield UL, that is VaR. The two are complements, not substitutes.

10. Mark-to-Market: Migration Risk and Spread Risk

Default Mode assumes loss happens only on default. Mark-to-Market recognises that value changes with market perception even while the exposure is still performing. Two new risk profiles appear.

- **Migration risk**: loss in value due to a rating downgrade. Mechanism: the downgrade forces the market to require a higher credit spread; the bond's contractual cash flows are fixed, so its price must fall to offer the higher yield. Price = Sum over k of C/(1+r)^k + FV/(1+r)^N, with r = risk-free rate + spread.
- **Spread risk**: loss in value from a generalised increase in market-required credit spreads, even with the borrower's own rating unchanged. This typically happens under market stress: a higher market price of risk lifts spreads for all risk levels, and existing instruments contractually bound to lower returns lose value.

Why bother with MTM at all? It captures risks Default Mode misses, gives a more dynamic view of value, and matters for marketable credit instruments such as bonds and securitised loans; since almost all loans can be securitised, it is relevant to almost every asset class, and useful for internal decisions even where loan accounting is not strict MTM.

11. Pro-cyclicality and the Role of Ratings

MTM has a recognised drawback: pro-cyclicality. In a downturn, credit risk rises, MTM asset values fall through migration and spread effects, bank capital shrinks with asset values, lending capacity contracts, and the economic cycle worsens. This feedback loop is a key debate in financial regulation.

Ratings feed both paradigms. In Default Mode models, the probability distributions of default risk and severity risk show higher variability the worse the ratings are, and they drive default-driven UL. In MTM portfolio models, ratings additionally generate migration matrices and credit spread curves, driving value-change-driven UL, default included.

12. Must Know Absolutely

Must know:
• EL = PD x LGD x EAD; LGD = 1 - RR, estimated by actual payouts (discounted net recoveries over EAD) or by market prices one month post-default.
• Aggregation: line ELR = PD x LGD; borrower ELR weights line LGDs by EAD shares; portfolio ELR weights borrowers by EAD. PD belongs to the borrower, LGD to the line.
• EAD = Em x Ec; Em > 1 for discretionary facilities, Em < 1 for amortizing loans.
• Basel II default: more than 90 days past due or Unlikely to Pay; wider and more objective than the traditional bad-loans definition.
• EL is covered by provisions; UL is variability around EL and requires capital.
• Credit loss distributions are skewed, so sigma fails and VaR measures UL: at 99% confidence, losses stay below EL + VaR with 99% probability; VaR is economic capital.
• EL is additive, VaR is not: diversification through imperfect correlation; concentration risk raises portfolio VaR; marginal VaR = VaR(p+i) - VaR(p).
• SCRM: default mode, stand-alone VaR, no correlations. FPCRM: portfolio and marginal VaR with correlations, default mode or MTM.
• MTM adds migration risk (downgrade, higher spread, price falls) and spread risk (general spread widening), at the cost of pro-cyclicality.